Noah Lee

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College Station, TX
Available for consulting and contract work

Summary

Computer vision engineer with four years building production ML systems across defense, maritime, and medical imaging. I take projects end to end: I design the models, set up the training and evaluation pipeline, make them fast enough for the hardware they run on, and see the deployment through. I work comfortably with external partners and am good at turning an ML problem into something that actually holds up in the field. Available for project-based work in computer vision, model deployment, and edge and embedded ML.

Selected Projects

- **Marine Perception for Autonomous Surface Vessels** — MIT Lincoln Laboratory / Confidential Partner
  Perception Lead — April - August 2026
  - Led computer vision for autonomous surface vessels, owning perception end to end from model design through field deployment.
  - Built and shipped perception models on visible and thermal imagery that run in real time on edge hardware for maritime operations.
  - Delivered a lightweight detection system for wide-scale aerial search and rescue that runs in real time on CPU.
  - Architected GPU-accelerated deployment so video streams are decoded and scored in memory, keeping latency low on live feeds.

Professional Experience

- **MIT Lincoln Laboratory** — Lexington, MA
  Technical Staff, Machine Learning — July 2022 - Present
  - Built multimodal foundation-model systems for long-term action recognition on body-worn camera footage, including structured generation to label unlabeled video at scale.
  - Benchmarked vision-language foundation models on satellite hyperspectral imagery under limited data and published the scaling-law analysis; extended the same approach to open-set detection in X-ray imagery using natural language.
  - Delivered 3D segmentation pipelines that detect hazardous materials in luggage at security checkpoints, lifting 2D detections into 3D volumes and deploying on a multi-node cluster for low-latency inference.
  - Owned datasets and experiment pipelines for a satellite remote-sensing program and introduced CI workflows for reproducible experiments.
  - Built microservice pipelines that enable self-service screening in real time at airport checkpoints, including finetuned anomaly detection on radar and a self-supervised pose model trained from millimeter-wave returns.
- **Massachusetts Institute of Technology** — Cambridge, MA
  Research Assistant — January 2023 - September 2023
  DAF-MIT AI Accelerator - Pilot Performance Prediction Using Physiological Metrics
  - Built algorithms to estimate pilot cognitive stress during difficult landings from physiological and behavioral signals (EMG, ECG, gaze).
  - Synchronized multimodal timeseries and loaded 140 GB of data in five minutes via dynamic loading.
  - Benchmarked TSFresh against MINIROCKET architectures for scalability and accuracy and published the results.
- **The MITRE Corporation** — McLean, VA
  Systems Engineer (Co-op) — May 2021 - December 2021
  - Used probabilistic graphical models of radar tracks to detect and classify air traffic in low-visibility conditions.
  - Simulated air traffic controllers with executable activity diagrams to surface potential mistakes and efficiency improvements.

Teaching & Mentoring

- **MIT Lincoln Laboratory - SOCOM Ignite**
  AI Workshop Instructor & Data Fusion Team Mentor 2022 - 2024
  - Ran an introductory AI workshop and mentored a student data fusion team, explaining complex ML concepts to non-specialists.
- **Northeastern University**
  Teaching Assistant, Machine Learning & Data Mining; CS Fundamentals 2021 - 2022
  - Supported undergraduate machine learning and core CS courses.

Technical Skills

- **Languages:** Python, C++, MATLAB, Java
- **Machine Learning & Vision:** PyTorch, JAX, TensorFlow/Keras, ONNX
- **Deployment & Infrastructure:** Docker, Redis, Slurm, git, CI/CD, GPU-accelerated and edge deployment

Education

Texas A&M University College Station, TX
M.S. in Computer Engineering (MSCE) Fall 2026 - Expected Spring 2028
Thesis: vision-language models and computer vision to interpret and vectorize construction blueprint PDFs.

Northeastern University, Khoury College of Computer Sciences Boston, MA
Bachelors of Science in Computer Science September 2019 - May 2022
Summa Cum Laude, Dean's List **GPA: 3.96/4.00**

Publications

- **N. Lee**, P. W. Moore and L. J. Brattain, "Scalable Deep Learning for Pilot Performance Analysis Using Multimodal Physiological Time Series," 2023 IEEE High Performance Extreme Computing Conference, doi: 10.1109/HPEC58863.2023.10363513.
- K. Vogt-Lowell*, **N. Lee***, T. Tsiligkaridis and M. Vaillant (* equal contribution), "Robust Fine-Tuning of Vision-Language Models for Domain Generalization," 2023 IEEE High Performance Extreme Computing Conference, doi: 10.1109/HPEC58863.2023.10363450.

Awards

MIT Lincoln Laboratory 2023-2024 Team Award July 2024